

An impact measure for news: its use in daily trading strategies

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Abstract

We investigate how 'news sentiment' in general and the 'impact of news' in particular can be utilised in designing equity trading strategies. News is an event that moves the market in a small way or a big way. We have introduced a derived measure of news impact score which takes into consideration news flow and decay of sentiment. Since asset behaviour is characterised by return, volatility and liquidity we first consider a predictive analytic model in which market data and impact scores are the inputs and also the independent variables of the model. We finally describe the trading strategies which take into consideration the three important characteristics of an asset, namely, return, volatility and liquidity. The minute-bar market data as well as intraday news sentiment metadata have been provided by Thomson Reuters.

Keywords: news sentiment, news impact, efficient market hypothesis daily trading , predictive analytics, asset returns, volatility of assets, liquidity of assets, GARCH model , bid-ask spread , news metadata , market data, volatility pumping, Kelly strategy.

1 Introduction

1.1 Why sentiment analysis

A major plank in the development and application of behavioural finance is the consideration of bounded rationality as introduced by Nobel laureate Herb Simon (Simon, 1964). It follows from the theme of bounded rationality and later works of behavioural theorists/economists, Kahneman and Tverskey, Sheffrin, Shiller that human beings in general and retail investors in particular are influenced by various psychological imperatives of 'fear, greed and exuberance'. This is in sharp contrast with the postulates of neoclassical theories of rational economic behaviour and scrupulous application of logic in decision making. Thus behavioural finance in many ways determine the risk attitudes and the investment goals of the (high net worth) individuals: so called HNIs who account for the majority of the invested wealth.

The neoclassical finance theory had embraced the efficient market hypothesis (EMH) which has become the cornerstone of investment science. The doubts about EMH and recent surge of interest in behavioral finance (Kahneman and Tversky, 1979, Kahneman, 2002, Sheffrin, 2008) not only opened the debate and exposed the limits of EMH but also reinforced the important role of sentiment and investor psychology in market behaviour. The central tenet of EMH is information arrival, that is, news is rapidly digested by rational stakeholders; yet the work of Shiller (2000), Hais (2010), and Barber and Odean (2008) reinforce the irrational contrarian and herd behaviour of the investors. Today the availability of sophisticated computer systems facilitating high frequency trading (Goodhart and

O'Hara, 1997) as well as access to automated analysis of news feeds (Tetlock, 2007, Mitra and Mitra, 2011) set the backdrop for computer automated trading which is enhanced by news. How investment strategies may harness sentiment of news events and also that of the market continue to be actively studied and reported by researchers in the investment community (Kahn, Dion, Brar, Hafez). Also see "Inside the Investor's Brain" (Wiley, 2007) by Richard Peterson. In this study we investigate how 'news sentiment' in general and the 'impact of news' in particular can be utilised in designing equity trading strategies.

1.2 Characterising Asset Behaviour

The majority of research on asset behaviour has focused on analysing historical market data and the construction of models which fit the data and best represent the true information held in such datasets. The key features studied in these models are return and volatility of stock price, which are considered in the decision making stage of trading. In this report, however, we introduce two further (novel) concepts with a view to enhance these classical methods. Firstly, our model looks at three characteristics of an asset, namely return, volatility, and liquidity. The consideration of liquidity, the additional feature, provides knowledge on the condition of markets and more importantly indicates whether a profitable signal can be successfully executed or not. Simply being able to determine a profitable position in the markets, through observation of stock price return and volatility, is not sufficient if in fact the actual trade is not available. Secondly, the investigational data set is expanded with the incorporation of news sentiment metadata in addition to the historical market data. The data sets are fused together to form two intraday time series on a minute bar scale. This data fusion enables us to construct an enhanced predictive analytics model for three parameters which characterise asset price behaviour.

The asset price parameters have been traditionally represented by two methods: predictive modelling and explanatory modelling. One of the most well-known explanatory models is the factor model (Fama and French, 1992). They capture return by extending the Capital Asset Pricing Model (CAPM), which was independently proposed by Treynor (1961), Sharpe (1964), Lintner (1965), and Mossin (1966), to include market capitalisation size and book-to-market ratio as explanatory factors. Such types of models can be categorised under 3 groups, that is macroeconomic, fundamental and statistical factor models depending on the choice and nature of these factors and how the respective models are calibrated. Although factor models have dominated this field of finance, a weakness is in their failure to quickly update changes in market conditions. The structure of these models is only single period inhibiting the incorporation of relevant past information at a sufficient speed. Parameters are updated through calibration but only at a slow pace where the model adapts. Mitra, Mitra and diBartolomeo (2009) have shown how the incorporation of news enhances the results of factor models and leads to an early prediction of changes in volatility. We believe that news sentiment should also be considered as an explanatory factor in the pricing of assets and therefore return. However, our approach turns to another class of

models that better predict volatility in a time-varying framework – generalised autoregressive conditional heteroskedasticity (GARCH) models.

By incorporating the additional factor of news sentiment in GARCH models, the near term volatility can be estimated using past news events over many periods. This near term estimation may be obtained using a number of alternative predictive models. Established models in this class are linear regression models (Stambaugh, 1999, Robertson and Wright, 2009), autoregressive (AR) models, moving average (MA) models and generalised autoregressive conditional heteroskedasticity (GARCH) (Bollerslev, 1986) models for volatility in particular. Our work utilises the family of AR and GARCH models within which news sentiment is introduced as an exogenous variable. Among the many applications of predictive modelling, the construction of preliminary stages to scenario generation is the most common practice. In the context of financial assets, scenario generators for the possible movements in stock price are formed from a predictive model for stock price.

Research studies which address the relationship between stock price movements and news sentiment have been growing in recent years (Tetlock, 2007, Barber and Odean 2008, Mitra, Mitra and diBartolomeo 2009, Leinweber and Sisk, 2009, Sinha, 2011). The earliest work that incorporates news articles in the influence of stock prices dates back to Niederhoffer (1971), where the font size of news print was used to determine the relative importance of news events and measure how they affected the stock market. Later, news was categorised into scheduled and non-scheduled news with researchers exploring their respective effects on stock returns and volatility (Lee, 1992, Engle, 1998). Tetlock (2007) was one of the first to use merely news sentiment data as a comparison to the movement in stock prices and trading activity.

1.3 Role of Liquidity in Asset Behaviour

Traditionally research on asset behaviour has focused on the study of two main features, namely, stock price and volatility of stock prices. Since the financial crisis of 2008 with the collapse of Lehman Brothers and the subsequent banking crisis and bank bail outs in the UK and other European countries the importance of liquidity has become paramount and is widely acknowledged by the finance community. Another implication of the financial crisis has been that regulators have imposed stricter liquidity requirements for banks, especially those active in the low latency trading activities (Gomber et al. 2011, AFM 2010). So for trading activities, the availability of liquidity to traders and brokers is of paramount importance. Hence many of the recent studies involving impact of news consider the implication on liquidity. For instance, Gross-Klaussman and Hautsch (2011) use the bid-ask spread, trading volume and market depth as proxies for liquidity with results showing greater increase of bid-ask spreads during news releases as opposed to market depth which does not differ much. Furthermore, Riordan, Storkenmaier et al. (2013) found that liquidity increases with news releases that have positive or neutral sentiment whereas negative news sentiment gives a corresponding decrease in liquidity.

It is well accepted in the trading community that liquidity has the role of monitoring market conditions and assessing the viability of orders decided by trading algorithms by taking into consideration the spread and the depth of the market. Given that margin requirement is an important and defining aspect of trading it can be argued that liquidity is an important determinant in deriving trading strategies and should be introduced as a parameter in predictive models which characterise asset behaviour. There are a variety of definitions which explain the role and measure of liquidity mainly in the context of trading. Spread measures view liquidity from the point of the cost that one has to bear for immediate trade, in other words the viability of orders, and typical measures include the effective spread and the bid-ask spread. Depth measures consider liquidity as the effect of large orders on a particular asset hence looking at market conditions, and are often measured using traded volume, order volume or Kyle's λ (Kyle, 1985). A short but comprehensive discussion of liquidity measures is given in chapter ## of Mitra, Yu et al (2015).

1.4 Modelling Architecture and Choice of Assets

In our investigation, two different datasets are used. We (i) consider the predictive models and investigate whether the additional information stream of news metadata improves the results of the predictive models. A selection of five assets from the Finance industry is used to test the models in an intraday setting for the year 2008. We then (ii) use the results of the predictive model to construct a daily trading strategy which uses the 'Kelly Criterion'. As we have already restated the earlier findings of one of the authors, Mitra and Mitra (2011), using news metadata leads to earlier and superior prediction of volatility. Kelly Criterion uses volatility (see volatility pumping by Luenberger (1997)) to compute the 'trading bets'; therefore we use this approach to determine our daily equity trading strategy. The index FTSE100 and its constituents are considered in our investigation for the years 2013 and 2014. The experimental platform can be easily adapted for higher frequency lower latency trading. It can be equally adapted for automated trading of other asset classes.

1.5 Outline of the paper

The rest of the paper is organised as follows: In section 2 we discuss the data instances that are used in our study. In section 3 we describe the key measures which we have used. First the return, volatility and liquidity measures are introduced. Then we define sentiment measures and in particular the positive and negative sentiment measures. Using accumulation and decay of sentiments we derive the positive and negative impact scores. Finally we introduce an approximation whereby we determine the number of past news items we may need to consider. In section 4 we set out the predictive models which describe price, volatility and liquidity of assets. Computational results for the predictive volatility model are also given. In section 5 we describe a daily trading strategy which is based on the Kelly's betting formula. In this section we also report the

results of applying the strategy to equities taken from FTSE100 index. A summary of our findings and conclusions are presented in section 6.

2 Data

Our modelling architecture uses two streams of time series data: (i) The market data which is given at the minute bar level as well as daily frequency, and includes bid price, ask price and the execution price, (ii) News metadata as supplied by Thomson Reuters.

2.1 Market Data

Both daily and intraday prices are used in this chapter, for testing the trading strategy and predictive models respectively. The data fields of the market data are set out in table 2.1. The trading strategy (in section 5) is tested on the entire asset universe of FTSE100.

<i>Data Field</i>	<i>Field Name</i>	<i>Description</i>
1	##RIC	Reuters instrument code individually assigned to each company.
2	Date	In the format DD-MM-YYYY.
3	Time	In the format hh:mm:ss, given to the nearest minute.
4	GMT Offset	Difference from Greenwich Mean Time.
5	Type	Type of market data – in this case “Intraday 1 min”
6	Last	Last prices for the corresponding minute
7	Close Bid	Close bid prices for the corresponding minute
8	Close Ask	Close ask prices for the corresponding minute

Table 2.1: Description of all the data fields for a company in the market data.

The high frequency intraday market data, used for the predictive models, is compiled on a minute-bar scale for a selection of 5 assets from the Finance sector (AIG, Barclays, Bank of America, HSBC and JP Morgan). These companies have been chosen because of their large market capitalisation, which guarantees a wide coverage of news. As a consequence, we are able to generate a sufficient number of data points in the time series of news metadata.

We extract the data from 08:00-18:30 each day to make up a **trading day** such that pre-trade and post-trade hours are included; thus any news event (and the corresponding sentiment) captured outside of trading hours is incorporated into our predictive model and any loss of information is avoided (impact of news).

2.2 Sentiment and News Metadata

News analytics data is presented in the well established metadata format whereby a news event is given tags of relevance, novelty and sentiment (scores) for a given individual asset. The analytical process of computing such scores is fully automated from collecting, extracting, aggregating, to categorising and scoring. The result is an individual score assigned to each news article for each characteristic using scales from 0-100 or probabilities. Although we use intraday news data, for a given asset the number of news stories, hence data points are variable and do not match the time frequencies of market data. The attributes of news stories used in our study are Relevance and Sentiment.

The news metadata for the chosen assets were selected under the filter of relevance score, that is, any news item that had a relevance score under the value of 70 was ignored and not included in the data set. This ensured with a high degree of certainty that the sentiment scores to be used are indeed focused on the chosen asset and is not just a mention in the news for comparison purposes for example.

The final preparation task for the data is to align the frequency of sentiment scores to the trading hours of 08:00-18:30. Any news item released before or after these trading times were summed and bucketed in the next time period, which may be the following day. As a consequence, there is no discarding of news sentiment which could be influencing the price and return thus, not a single piece of news data is ignored.

3 Measures: Return, Volatility, Liquidity, Sentiment and Impact

3.1 Return, Volatility and Liquidity Measures

With the raw market data that is provided to us, we are able to calculate asset behaviour measures using close prices, bid prices and ask prices.

Bucket Size = 1 minute; Data Frequency = Minute Bar.

The **trading day** starts at 08:00 hours and ends at 18:30 hours thus in a trading day the total number of buckets is 630.

Any news and sentiment retrieved overnight (between 18:31 – 07:59 the following day) is put in the first time bucket of the following day i.e. 08:00. The same applies to weekend news, which is aggregated over both days and considered in the first minute when trading is resumed. This method of categorisation reflects our belief that reactions to news are reflected in stock price movements immediately after release or even days after.

Return

The return measure that we use in the model is the log-return calculated by the following equation

$$R_t = \text{Log}(P_t) - \text{Log}(P_{t-1}) \quad [3.1]$$

where P_t is the close price at time t .

Volatility

The volatility measure used in the model is calculated as a rolling standard deviation of log returns for one **trading day** (see section 2.3) leading to 630 data points.

Liquidity

Liquidity is represented by the spread of the bid and ask prices, which measures the cost one has to bear for immediate trade. Equation 3.2 gives the expression for bid-ask spread.

$$\text{Spread} = \frac{\text{Ask} - \text{Bid}}{\text{Mid}} \quad [3.2]$$

where Mid is the mid-price between the bid and ask prices.

3.2 Sentiment Measure

In our analytical model we have introduced two concepts, namely, (i) sentiment score and (ii) impact score. The sentiment score is a quantification of the mood (of a typical investor) in respect of a news event. The impact score takes into consideration the decay of the sentiment of one or more news events and how after aggregation these impact the asset behaviour.

Sentiment Score

Thomson Reuters' news sentiment engine analyses and processes each news story that arrives as a machine readable text. Through text analysis and other classification schemes the engine then computes for each news event: (i) relevance, (ii) entity recognition and (iii) sentiment probabilities as well as a few other attributes (see Mitra and Mitra, 2011). A news event sentiment can be positive, neutral and negative and the classifier assigns probabilities such that

$$\text{Prob}(\text{positive}) + \text{Prob}(\text{neutral}) + \text{Prob}(\text{negative}) = 1.0 \quad [3.3]$$

We turn these three probabilities into a single sentiment score in the range +50 to -50 using the following equation:

$$Sent = 100 * \left(Prob(positive) + \frac{1}{2} Prob(neutral) \right) - 50 \quad [3.4]$$

where *Sent* denotes a single transformed sentiment score. We find that such a derived single score provides a relatively better interpretation of the mood of the news item. Thus the news sentiment score is a relative number which describes the degree of positivity and negativity in a piece of news. During the trading day, as news arrives it is given a sentiment value. Given that

$$-50 \leq Sent \leq 50$$

for a given news item k at the time bucket t_k , we define $PNews(k, t_k)$ and $NNews(k, t_k)$ as the sentiment of the k^{th} news (see the following section).

3.3 Impact Score

It is well known from research studies that news flow affects asset behaviour (Patton and Verardo, 2012, Mitra, Mitra and diBartolomeo, 2009). Therefore, accumulation of news items as they arrive is important. Patton and Verardo (2012) noticed decay in the impact of news on asset prices and their betas on a daily timescale and further determine the complete disappearance of news effects within 2-5 days. It was also observed in Arbex-Valle, Erlwein-Sayer et al. (2013). Mitra, Mitra and diBartolomeo (2009) created a composite sentiment score in their volatility models after initial experiments revealed no effect on volatility predictions with sentiment alone; the decay period in this study was over 7 days.

In order to compute the impact of news events over time we first find an expression which describes the attenuation of the news sentiment score. The impact of a news item does not solely have an effect on the markets at the time of release; the impact also persists over finite periods of time that follow. To account for this prolonged impact, we have applied an attenuation technique to reflect the instantaneous impact of news releases and the decay of this impact over a subsequent period of time. The technique combines exponential decay and accumulation of the sentiment score over a given time bucket under observation. We take into consideration the attenuation of positive sentiment to the neutral value and the rise of negative sentiment also to the neutral value and accumulate (sum) these sentiment scores separately. The separation of the positive and negative sentiment scores is only logical as this avoids cancellation effects. For instance cancellation reduces the news flow and an exact cancellation leads to the misinterpretation of no news.

News arrives asynchronously; depending on the nature of sentiment it creates, we classify these into three categories, namely, positive, neutral and negative. For the purpose of deriving impact measures, we only consider the positive and negative news items.

Let

POS denote the set of news with positive sentiment value $Sent > 0$;

NEG denote the set of news with negative sentiment value $Sent < 0$;

and

$PNews(k, t_k)$ denote the sentiment value of the k^{th} positive news arriving at time bucket t_k , $1 \leq t_k \leq 630$ and $k \in POS$; $PNews(k, t_k) > 0$.

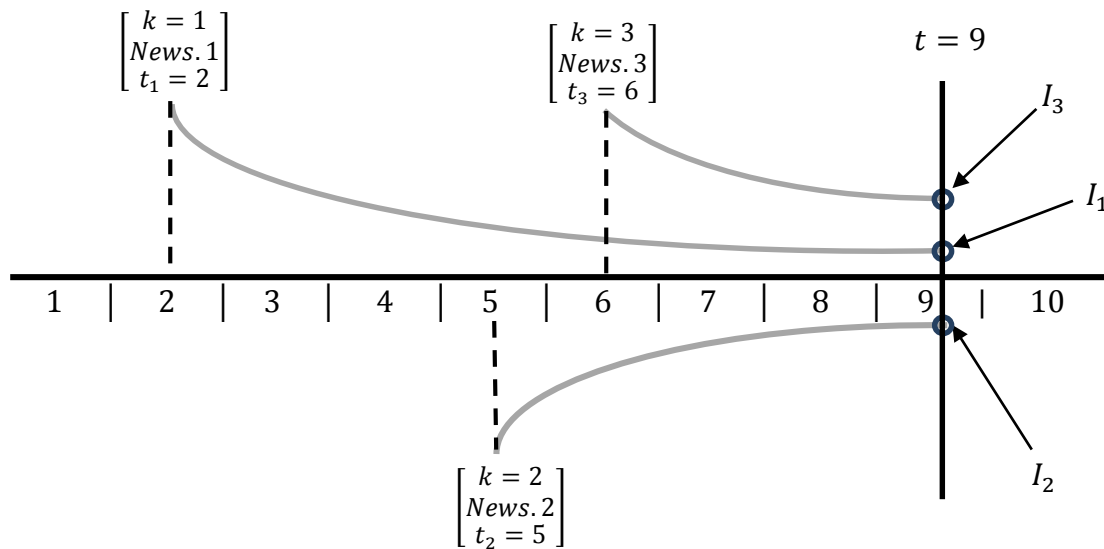
$NNews(k, t_k)$ denote the sentiment value of the k^{th} negative news arriving at time bucket t_k , $1 \leq t_k \leq 630$ and $k \in NEG$; $NNews(k, t_k) < 0$.

Let λ denote the exponent which determines the decay rate. We have chosen λ such that the sentiment value decays to half the initial value in a 90 minute time span. The cumulated positive and negative sentiment scores for one day are calculated as:

$$PImpact(t) = \sum_{\substack{k \in POS \\ t_k \leq t}} PNews(k, t_k) e^{-\lambda(t-1)} \quad [3.6]$$

$$NImpact(t) = \sum_{\substack{k \in NEG \\ t_k \leq t}} NNews(k, t_k) e^{-\lambda(t-1)} \quad [3.7]$$

In equations 3.6 and 3.7 for intraday $PImpact$ and $NImpact$, t is in the range, $t = 1, \dots, 630$. On the other hand for a given asset all the relevant news items which arrived in the past, in principle, have an impact for the asset. Hence, the range of t can be widened to consider past news, that is, news which are 2 or more days 'old'. We deal with this contingency by an approximation which is described in section 3.4. In figure 3.1 we provide an illustration of how $PImpact(9, \lambda)$ and $NImpact(9, \lambda)$ are computed for the two positive news items (News.1 and News.3) and the negative news item (News.2) respectively.



$$\begin{aligned}
 \text{Impact of News.1} &= I_1 = P\text{News}(1,2)e^{-\lambda(9-2)} \\
 \text{Impact of News.2} &= I_2 = N\text{News}(2,5)e^{-\lambda(9-5)} \\
 \text{Impact of News.3} &= I_3 = P\text{News}(3,6)e^{-\lambda(9-6)}
 \end{aligned}$$

Figure 3.1: An illustration of how impact scores are calculated for individual news items that have positive or negative sentiment.

The arrival of more news items leads to higher values of accumulation; this therefore takes into account the news intensity, that is, the news flow. The impact scores for JP Morgan during the month of August 2008 is illustrated in figure 3.2 showing the positive and negative scores (with attenuation and accumulation).

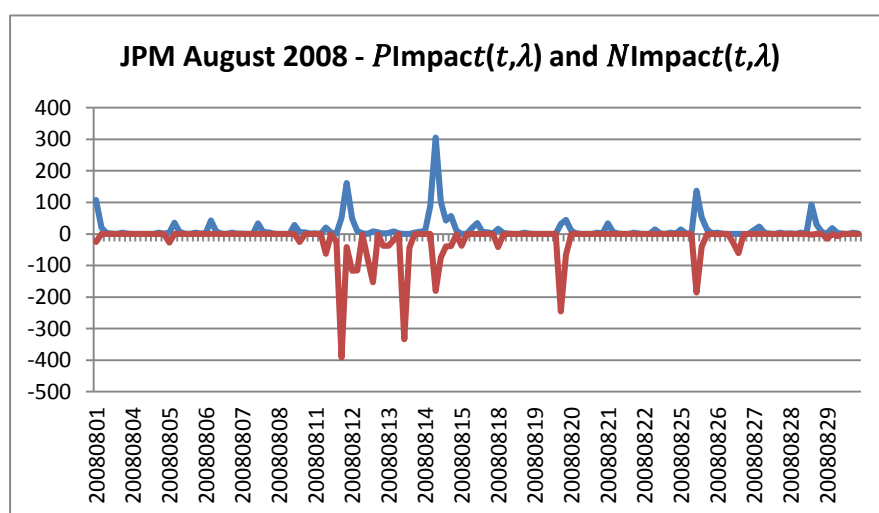


Figure 3.2: JP Morgan August 2008: News impact score (accumulated and aggregated) for positive (blue line) and negative (red line) sentiment respectively.

Moreover, the actual number of news volume across each time period can also be incorporated into the impact score, or a scaled down measure of news volume. This impact measure is flexible to any number of variables being added to it, examples being news categories or market capitalisations. By incorporating these variables, more features about the news item and the particular asset are fed into the impact score to create a more informative measure.

3.4 Look Back Period: An Approximation

As set out in section 3.3, for a given asset all the past news items have an impact for the asset. Therefore, we define $T_{LBP}(t)$ as the number of past time buckets we need to consider looking back from the current time bucket t , such that the corresponding $PImpact$ and $NImpact$ computations will be an acceptable approximation controlled by a threshold parameter θ . θ is the threshold expressed as the sentiment value that is considered large enough for inclusion in the impact computation for a given asset. This concept is illustrated in figure 3.3. The look back period (LBP) is defined as the number of time buckets T_{LBP} .

Let

$$t_{LBP}^k = \min(t - t_k) \quad [3.8]$$

where $PNews(k, t_k)e^{-\lambda(t-t_k)} \leq \theta$.

Therefore T_{LBP}^k is given by the expression

$$T_{LBP}^k = \max_k \{t_{LBP}^k\} \quad [3.9]$$

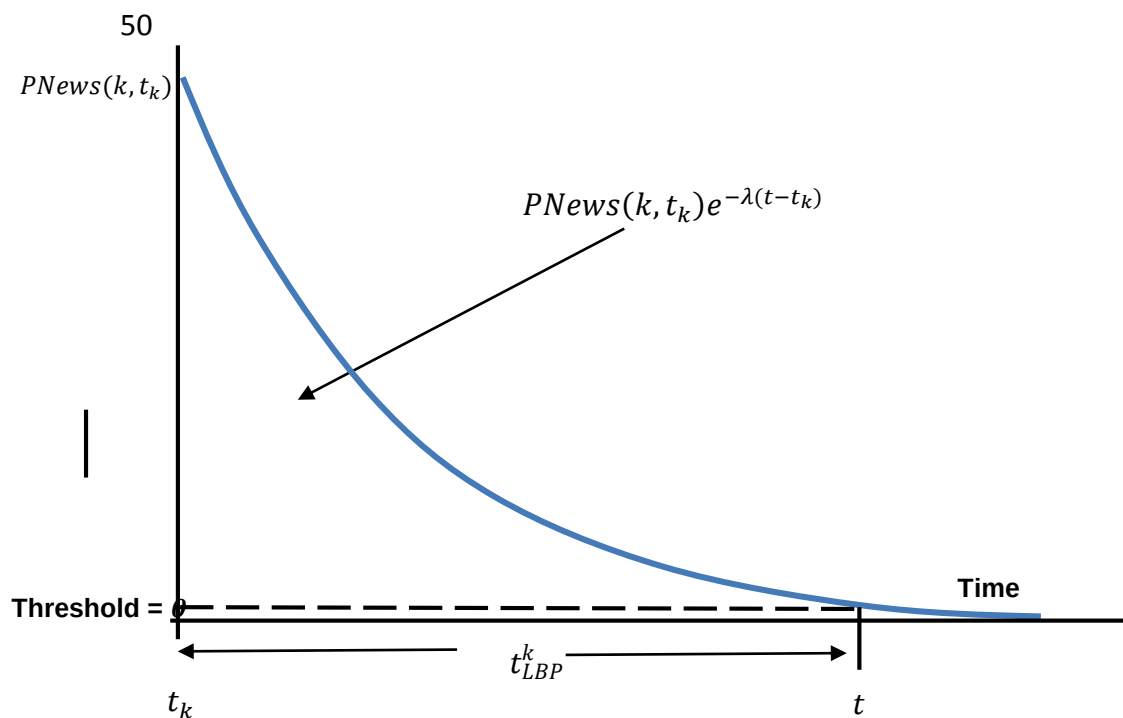


Figure 3.3: t_{LBP}^k illustrated using a positive news item.

In the case of negative sentiment value, the condition with respect to θ then becomes $NNews(k, t_k)e^{-\lambda(t-t_k)} \geq \theta$. So considering all the assets $i = 1, 2, \dots, N$ in a given index we derive the LBP for the index as:

$$T_{LBP}^* = \max_i \{T_{LBP}(i)\} \quad [3.10]$$

where $T_{LBP}(i)$ is the LBP for the asset i .

4 The Model

4.1 Autoregressive and GARCH Models

Predictive analytical models for stock price return and volatility, such as the Autoregressive Conditional Heteroskedasticity ARCH model (Engle, 1982) and Generalised Autoregressive Conditional Heteroskedasticity GARCH model (Bollerslev, 1986), are well understood and applied extensively. These models exploit the techniques of time series analysis (Chatfield, 2009; Harvey, 1990) and are able to account for the correlation between each of the individual points in the time series of stock price returns by fitting them to models which are of Autoregressive (AR) nature. We have adopted this approach in the construction of our models for predicting return, volatility, and bid-ask spread. However, our method differs from ARCH/GARCH type models as it deals with two time series, (i) the first time series is that of market data and (ii) the second time series is the news metadata. This approach can be seen as an extension of GARCH and AR models with the variable of news sentiment being the innovative addition. Another feature that distinguishes this from existing predictive models is the joint modelling of three key variables which describe asset behaviour, namely, return, volatility and liquidity, whereas the majority of existing literature focuses only on one variable, either return or volatility (Kothari, Zimmerman 1995; French, Schwert, Stambaugh 1987). We believe that apart from the classic measures of return and volatility that needs to be evaluated and monitored for risk control and trading strategies, it is also now imperative to include a liquidity measure as this reflects the conditions of the markets and feasibility of trades. This can be determined from two perspectives: (i) the spread of the market, evaluated by the bid-ask spread, and (ii) the depth of the market, computed by the total volume of bids and asks. We set out below a detailed description of the model:

Asset Return

Taking into consideration the inclusion of two time series, we construct an AR(2) predictive model for return with the enhancement of a news impact score (see section 3.3).

Let, R_t be the log return of stock prices at time period t

R_{t-i} be the log return of stock prices by lag i

$PImpact_{t-1}$ be the positive news impact score of the previous time interval

$NImpact_{t-1}$ be the negative news impact score of the previous time interval

θ_i be the weighting coefficients to be estimated
 e_t be the error term

Therefore, the log-return R_t is given by the expression,

$$R_t = \theta_0 + \theta_1 R_{t-1} + \theta_2 R_{t-2} + \theta_3 PImpact_{t-1} + \theta_4 NImpact_{t-1} + e_t \quad [4.1]$$

Asset Volatility

The prediction model for volatility is an extended GARCH(1,1) model, as before with the addition of a news impact score.

Let, σ_t^2 be the squared volatility at time t
 ϵ_{t-1}^2 be the lagged log-return residuals
 σ_{t-1}^2 be the lagged squared volatility
 $PImpact_{t-1}$ be the positive news impact score of the previous time interval
 $NImpact_{t-1}$ be the negative news impact score of the previous time interval
 $\alpha_i, \beta_1, \omega_i$ be the weighting coefficients to be estimated
 u_t be the error term

Therefore, σ_t^2 is given by the expression,

$$\sigma_t^2 = \alpha_0 + \alpha_1 \epsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2 + \omega_1 PImpact_{t-1} + \omega_2 NImpact_{t-1} + u_t \quad [4.2]$$

Asset Liquidity

We measure the liquidity of the chosen asset by the bid-ask spread by constructing an AR(3) model with the addition of the news impact score.

Let, S_t be the bid-ask spread at time t
 S_{t-i} be the bid-ask spreads at lag i
 $PImpact_{t-1}$ be the positive news impact score of the previous time interval
 $NImpact_{t-1}$ be the negative news impact score of the previous time interval
 γ_i be the weighting coefficients to be estimated
 η_t be the error term

The model for S_t is as follows:

$$S_t = \gamma_0 + \gamma_1 S_{t-1} + \gamma_2 S_{t-2} + \gamma_3 S_{t-3} + \gamma_4 PImpact_{t-1} + \gamma_5 NImpact_{t-1} + \eta_t \quad [4.3]$$

4.2 Computational Results

Extensive test have been carried out on the predictive models for return, volatility and liquidity. Highly significant results were found for the extended GARCH model, some of which are displayed below (see table 4.1). The performance of the autoregressive models for return and liquidity is not substantial enough to be able to conclude that news sentiment certainly improves prediction. In this chapter, we only present results for volatility prediction because they are in line with later sections about trading strategies. In fact, the better performance of GARCH

enhanced with news sentiment is utilised to create a profitable trading strategy. For results on asset return and liquidity, see Yu et al. (2013).

From the fitting of the models it can be seen that news sentiment does indeed have an effect on the prediction of volatility as all coefficient estimates are non-zero. The small magnitude of news impact coefficients does not mean that they are insignificant. This is due to the large magnitude of news impact scores which have been aggregated over time.

<i>Asset</i>	<i>Parameter</i>	
AIG	α_0 : -6.921e-06	ω_1 : 1.126e-06
	α_1 : 5.325e-02	ω_2 : 6.311e-07
	β_1 : 8.994e-01	
Bank of America	α_0 : -5.652e-07	ω_1 : 1.021e-07
	α_1 : 5.072e-02	ω_2 : 1.625e-07
	β_1 : 8.997e-01	
Barclays	α_0 : -7.561e-07	ω_{1p} : 3.480e-07
	α_1 : 5.278e-02	ω_2 : 4.510e-07
	β_1 : 9.233e-01	
HSBC	α_0 : -4.655e-07	ω_1 : 1.329e-07
	α_1 : 4.667e-02	ω_2 : 1.753e-07
	β_1 : 9.278e-01	
JP Morgan	α_0 : -1.543e-06	ω_1 : 1.539e-07
	α_1 : 5.059e-02	ω_2 : 2.451e-08
	β_1 : 9.002e-01	

Table 5.1: Estimated coefficients for the 5 chosen assets.

To understand the extent to which news impact enhances the results of these models, we use the fitted model to predict out-of-sample values. The out-of-sample period is taken as January – March 2009. The conclusions interpreted from out-of-sample results are that news enhanced predictive models do better predict future values of liquidity and volatility but not for stock returns. Therefore, the inclusion of news sentiment to a predictive model, in the form of an impact score, does increase accuracy of predictions of volatility. Once again, only results for volatility are shown below. See Yu et al. (2013) for further results on liquidity and returns.

Asset	RMSE	MAE
AIG	$2.685e^{-3}$	$1.803e^{-3}$
Bank of America	$1.103e^{-3}$	$7.89e^{-4}$
Barclays	$1.419e^{-2}$	$9.89e^{-4}$
HSBC	$3.641e^{-3}$	$7.75e^{-4}$
JP Morgan	$6.431e^{-4}$	$5.11e^{-4}$

Table 5.2: Out-of-sample statistics for the 5 chosen assets.

As shown in figure 5.2, all assets performed better in the news enhanced GARCH model. Therefore, the inclusion of news sentiment as an exogenous variable to the univariate GARCH model significantly reduces prediction errors and estimates values far closer to the true values of volatility.

5 A Daily Trading Strategy

We use the results of the predictive models to define and test a daily trading strategy. We first set out the motivation as well as the rationale of our approach. As already reported, including news (impact) as an additional independent variable in the prediction of volatility significantly improves the accuracy in prediction. This conclusion is also reported by a number of other researchers (Patton and Verardo, 2012, Arbex-Valle, Erlwein-Sayer et al. 2013, Mitra, Mitra and diBartolomeo, 2009). We have therefore set out to explore the effectiveness of applying Kelly criterion which is based on harnessing volatility in deriving betting strategies. The underlying principles are based on the concepts (i) log optimal growth and (ii) volatility pumping, as explained by Luenberger (1997). The mathematical methods underpinning the approach are presented in section 5.1, along with an example, and the results of our prototype (experimental) investigation are presented in section 5.2.

The fact that news and the included sentiment improve the estimation of the mean and volatility of the asset returns can be employed in various ways. Here, we want to focus on the log-optimal investment strategy, based on the Kelly criterion; see MacLean et.al (2011) and Luenberger (1997).

5.1 Log return and Kelly Criterion

Kelly (1956) was one of the first to use logarithmic utility in gambling and repeated investment problems. The strategy maximizes the long run growth rate of the investment and is myopic in the sense that current investment decisions only depend on the current distribution of returns.

To understand the basic principle behind the strategy, consider the long run growth rate G which is defined as

$$G = \lim_{N \rightarrow \infty} \frac{1}{N} \log \left(\frac{W_N}{W_0} \right),$$

where W_0 is the initial wealth and W_N is the wealth after N successive investments. Let p and $(1-p)$ denote the corresponding success and failure probabilities of the bets (trades). Then the terminal wealth of an investor after N trades, is

$$W_N = W_0(1+f)^M(1-f)^{N-M},$$

where f is the constant fraction of wealth wagered in each bet (trade) and M is the number of games won. Combining the two expressions, we obtain

$$G = \lim_{N \rightarrow \infty} \frac{M}{N} \log(1+f) + \frac{N-M}{N} \log(1-f),$$

which approaches $G = p \log(1+f) + (1-p) \log(1-f)$ almost surely as n tends to infinity.

According to the independent nature of Bernoulli trials, maximizing G is hence equivalent to maximizing $E(W) = p \log(1+f) + (1-p) \log(1-f)$, that is, to maximize the expected logarithmic utility of each bet (trade).

The optimal fraction f^* is calculated to be $f^* = 2p-1$, which is the expected gain per trial. Note that if the payoff is $+b$ if the bet is won, the solution changes to $f^* = \frac{p(b+1)-1}{b}$.

In contrast to maximizing expected logarithmic utility, maximizing simply the expected value of the game, will lead to bankruptcy of the investor. This is because she is advised to invest all her capital in a game which is favourable. However, obviously, if this game is lost, all her money is gone.

Due to the constant fraction that is wagered in each game the absolute amount bet is monotone increasing in wealth and the investor never risks complete ruin. However, the fraction might be very large, especially if the wager is favourable and the risk is small. Besides, according to Thorp (1971), the Kelly portfolio does not necessarily lie on the efficient frontier in a mean-variance model.

5.2 Experimental Trading Results

We have investigated a daily trading strategy based on 'Kelly Criterion' for a subset of assets (equities) taken from the FTSE100 index for the two calendar years 2013 and 2014. The levels of news flow for the constituents of the FTSE100 were analysed and put on a ranking. The asset universe for the trading portfolio comprised of cash (at a risk free rate of 2%), FTSE100 index future and four assets, namely, Barclays, Vodafone, BP and HSBC. These four constituents were chosen

on the basis of ranked news flow intensity; they were also the most traded and had high capitalizations. In order to establish the advantages, or otherwise, of using news sentiment and news impact we have studied the results of applying this strategy for 'market data' only time series and 'market data' plus 'news metadata' time series.

In these experiments, rebalancing of the portfolio occurs daily, therefore, a new prediction of returns and volatility is computed every day using the previous 800 days (in-sample period) of daily returns and the corresponding 'news metadata'. The impact scores computed using the formula given in section 3.3 are incorporated in these predictions.

The FTSE100 index is chosen as the benchmark with which our trading strategies are compared. The period (2013-2014) for which we test the trading strategy is fairly lacklustre for FTSE100 index; it cannot be classified either as bull or bear regime. Two year return was a meagre 11%. Additionally, a major criticism of Kelly is that the corresponding strategy tends to "bet" large amounts (in relation to the trade portfolio value). This approach is not at all liked by "risk averse" investors. For this reason, a scaled down version is introduced, called partial Kelly. In the experiments we have compared and contrasted trade portfolios which

- (i) use full Kelly criterion with only market data
- (ii) use partial Kelly criterion with only market data
 - 50% partial Kelly
 - 30% partial Kelly
- (iii) use full Kelly criterion with market data and news impact measure
- (iv) use partial Kelly criterion with market data and news impact measure
 - 50% partial Kelly
 - 30% partial Kelly

The charts which display the behaviour of the benchmark and trading strategies (ii)...(v) are shown in figures 5.1 and 5.2. Risk averse approaches are covered with strategies (iii) and (v). The corresponding performance summaries are set out in table 5.1. Use of news sentiment and the corresponding benefit of accurate early prediction of volatility is easily seen from the displays and tables.

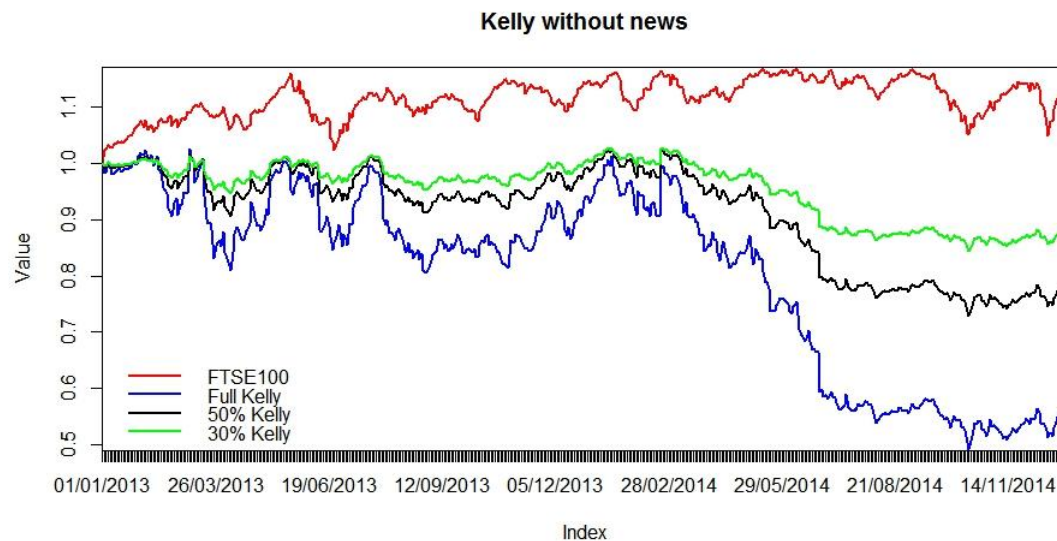


Figure 5.1: Performance of FTSE100, full Kelly criterion and partial Kelly criterion strategies without the incorporation of news impact measure for the years 2013-2014.

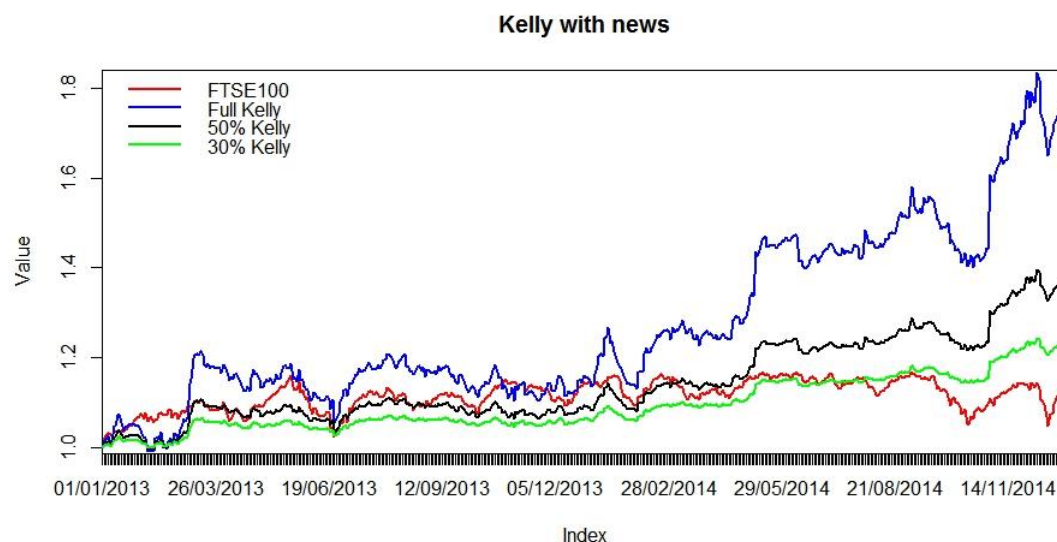


Figure 5.2: Performance of FTSE100, full Kelly criterion and partial Kelly criterion strategies with news impact measure for the years 2013-2014.

The mean and standard deviation of portfolio returns are presented for all seven strategies in table 5.1. Maximum drawdown is computed as the maximum peak-to-trough decline during any time of the whole period. The drawdown is quoted as the percentage between the peak and the trough. For the strategies that do not include news sentiment data, the corresponding portfolio performances are far worse in terms of profitability and risk. On the other hand, the set of strategies which incorporate news sentiment metadata not only beat the performance of the benchmark, but also give better values for down-side risk. In particular, the partial Kelly strategies with news display the most favourable properties in terms of Sharpe ratio, Sortino ratio and maximum drawdown. Although the full Kelly

strategy with news obtains the highest returns, due to the large positions suggested by the strategy, a high level of risk is also contained in this strategy.

Portfolio	Final Value	Mean returns	St. Dev. Returns	Excess RFR (%)	Sharpe Ratio	Sortino Ratio	Max draw-down (%)	Max. rec. Days
FTSE100	1.11	-	-	3.46	0.28	0.39	11.86	195
Full Kelly, no news	0.56	-0.00096	0.0186	-26.79	-1.03	-1.36	52.03	465
50% Kelly, no news	0.78	-0.00044	0.0093	-13.44	-0.96	-1.28	29.12	241
30% Kelly, no news	0.88	-0.00023	0.0056	-8.04	-0.93	-1.25	18.00	241
Full Kelly with news	1.73	0.00116	0.0134	28.99	1.19	1.99	13.16	213
50% Kelly with news	1.36	0.00062	0.0067	14.25	1.24	2.09	6.43	102
30% Kelly with news	1.22	0.00040	0.0040	8.48	1.25	2.13	3.67	102

Table 5.1: Summary statistics for all the strategies tested from 2013-2014.

6 Summary

Behavioural models and sentiment analysis are gaining momentum and acceptance within the investment community. Our study sets out to consider news sentiment and its impact on asset behaviour. We have introduced a novel concept of **news impact** which takes into consideration the volume of news and its decay over time. We have

(i) reported our study based on the predictive analysis of asset behaviour: return, volatility and liquidity. The main findings of this study are that by including news metadata as an additional time series and treated as an independent variable, the predictions of liquidity and volatility are improved, in particular, the latter. Given that **Kelly Criterion** uses asset volatility and that we are able to predict volatility early and accurately, we have created our daily trading strategy using Kelly's formula.

(ii) The FTSE100 assets and the FTSE100 index futures were investigated for the 'lacklustre' regime of 2013 and 2014. In these investigations we used (a) market data only and (b) market data plus news metadata. The results for the latter data set shows convincingly the superior performance when we use news metadata in addition to market data.

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7 References

1. Arbex-Valle, C., Erlwein-Sayer, C., Kochendörfer, A., Kübler, B., Mitra, G., Nzouankeu Nana, G.A., Nouwt, B. and Stalknecht, B. (2013). News-Enhanced Market Risk Management. Fraunhofer – OptiRisk whitepaper.
2. Barber, B. M., and Odean, T. (2008). All that glitters: The effect of attention and news on the buying behaviour of individual and institutional investors. *Review of Financial Studies*, 21(2), 785-818.
3. Bollerslev, T., Engle, R., and Wooldridge, J. (1988). A capital asset pricing model with time-varying covariances. *Journal of Political Economy*, 96(1), 116-131.
4. Bollerslev, T. (1986). Generalized Autoregressive Conditional Heteroskedasticity. *Journal of Econometrics*, 31(3), 307-327.
5. Brar, G. (2013). Text mining for longer horizon investors. [Presentation] *Behavioural Models and Sentiment Analysis Applied to Finance*, London 2 July. Available from: <http://unicom.co.uk/quant-finance/index.php>.
6. Chatfield, C. (2009). The Analysis of Time Series: An Introduction. *Sixth Edition, Chapman & Hall Texts in Statistical Science*.
7. Dion, M. (2013). Language recognition and news flow. [Presentation] *Behavioural Models and Sentiment Analysis Applied to Finance*, London 3 July. Available from: <http://unicom.co.uk/quant-finance/index.php>.
8. Engle, R. (1982). Autoregressive conditional heteroskedasticity with estimates of the variance of United Kingdom inflation. *Econometrica*, 50(4), 987-1007.
9. Engle, R. (1998). Macroeconomic announcements and volatility of treasury futures.
10. Fama, E. F. and French, K. R. (1992). The cross-section of expected stock returns. *Journal of Finance*, 47(2), 427-465.
11. French, K., Schwert, G.W., and Stambaugh, R.F. (1987). Expected stock returns and volatility. *Journal of Financial Economics*, 19, 3-29.
12. Goodhart, C.A., & O'Hara, M. (1997). High frequency data in financial markets: Issues and applications. *Journal of Empirical Finance*, 4(2), 73-114.
13. Groß-Klußmann, A., and Hautsch, N. (2011). When machines read the news: Using automated text analytics to quantify high frequency news-implied market reactions. *Journal of Empirical Finance*, 18(2), 321-340.

14. Hafez, P. (2013). Market-level Sentiment for trading Forex and equity indices. *[Presentation] Behavioural Models and Sentiment Analysis Applied to Finance, London 2 July*. Available from: <http://unicom.co.uk/quant-finance/index.php>.
15. Harvey, A.C. (1990). Forecasting structural time series models and the Kalman filter. *Cambridge University Press*.
16. Kahn, R. (2013). Quant 3.0: Harnessing the mood of the web in alpha strategies. *[Presentation] Behavioural Models and Sentiment Analysis Applied to Finance, London 3 July*. Available from: <http://unicom.co.uk/quant-finance/index.php>.
17. Kahneman, D. and Tversky, A. (1979). Prospect Theory: An Analysis of Decision under Risk. *Econometrica*, 47(2), 263 – 292.
18. Kahneman, D. (2002). Maps of bounded rationality: The [2002] Sveriges Riksbank Prize. *[Lecture] in Economic Sciences*. Link: http://nobelprize.org/nobel_prizes/economics/laureates/2002/kahneman-lecture.html
19. Kelly, J. L. (1956). A new interpretation of information rate. *Bell System Technical Journal*, 35, 917-926.
20. Kothari, S.P. and Zimmerman, J.L. (1995). Price and return models. *Journal of Accounting and Economics*, 20(2), 155-192.
21. Kyle, A. S. (1985). Continuous auctions and insider trading. *Econometrica: Journal of the Econometric Society*, 1315-1335.
22. Lattemann, C., Loos, P., Gomolka, J., Burghof, H. P., Breuer, A., Gomber, P., ... & Zajonz, R. (2012). High Frequency Trading. *Business & Information Systems Engineering*, 4(2), 93-108.
23. Lee, C.M.C. (1992). Earnings News and Small Traders. *Journal of Accounting and Economics*, 15, 265-302.
24. Leinweber, D., and Sisk, J. (2011). Relating news analytics to stock returns. Chapter 6 in Mitra, G. and Mitra, L. (2011). "The Handbook of News Analytics in Finance", *John Wiley & Sons*.
25. Lintner, J. (1965). The valuation of risky assets and the selection of risky investments in the portfolios and capital budgets. *Review of Economics and Statistics*, 47, 13-37.
26. Luenberger, D. (1997) Investment Science. *Oxford University Press*.
27. MacLean, L. C., Thorp, E. O. and Ziemba, W. T. (2011). The Kelly capital growth investment criterion: Theory and practice. *World Scientific*, 3.

28. Mitra, G., diBartolomeo, D., Banerjee, A. and Yu, X. (2011). Automated analysis of news to compute news sentiment: Its impact on liquidity and trading. Working paper. Available from: <http://optimrisk-systems.com/publications.asp#whitepaper>
29. Mitra, L., Mitra, G., and diBartolomeo, D., (2009). Equity portfolio risk (volatility) estimation using market information and sentiment. *Quantitative Finance*, 9(8), 887-895.
30. Mitra, L., and Mitra, G. (2011). Application of news analytics in finance: A review. Chapter 1 in Mitra, G. and Mitra, L. (2011) "The Handbook of News Analytics in Finance", *John Wiley & Sons*.
31. Mossin, J. (1966). Equilibrium in a capital asset market. *Econometrica: Journal of the Econometric Society*, 768-783.
32. Niederhoffer, V. (1971). The analysis of world events and stock prices. *The Journal of Business*, 44(2), 193-219.
33. Patton, A.J. and Verardo, M. (2012). Does beta move with news? Firm-specific information flows and learning about profitability. *Review of Financial Studies*, 25(9), 2789-2839.
34. Peterson, R.L. (2007). Inside the Investor's Brain. *John Wiley & Sons*.
35. Riordan, R., Storkenmaier, A., Wagener, M., & Sarah Zhang, S. (2013). Public information arrival: Price discovery and liquidity in electronic limit order markets. *Journal of Banking & Finance*, 37, 1148-1159.
36. Robertson, D., & Wright, S. (2009). *The Limits to Stock Return Predictability*. mimeo.
37. Sharpe, W. F. (1964). Capital asset prices: A theory of market equilibrium under conditions of risk. *Journal of finance*, 19(3), 425-442.
38. Shefrin, H. (2008). A Behavioral Approach to Asset Pricing. *Academic Press*.
39. Shiller, R. (2000). Irrational Exuberance. *Princeton University Press*.
40. Simon, H.A. (1964). On the concept of organizational goal. *Administrative Science Quarterly*, 1-22.
41. Sinha, N. (2010). Underreaction to news in the US stock market. Available at SSRN 1572614.
42. Stambaugh, R. F. (1999). Predictive regressions. *Journal of Financial Economics*, 54(3), 375-421.

43. Tetlock, P. C. (2007). Giving content to investor sentiment: The role of media in the stock market. *Journal of Finance*, 62(3), 1139-1168.
44. Thorp, E. O. (1971). Portfolio choice and the Kelly criterion. *Proceedings of the Business and Economics Section of the American Statistical Association*, 215-224.
45. Treynor, J.L. (1961). Toward a theory of market value of risky assets. Unpublished manuscript. A final version was published in "Asset Pricing and Portfolio Performance" (2009).
46. Yu, X., Mitra, G. and Yu, K. (2013). Impact of News on Asset Behaviour: Return, Volatility and Liquidity in an Intra-Day Setting. Available at SSRN: <http://ssrn.com/abstract=2296855>